PROJECT BRIEF

StudyPilot AI

Live assignment dashboard with an n8n-powered assistant. React shows the work; Vercel securely forwards API requests; n8n supplies the data and AI workflow.

LIVE APP
studypilot-one.vercel.app/

GITHUB REPOSITORY
github.com/milyaskhan-2127/StudyPilot

01 SYSTEM ARCHITECTURE

One route from browser to workflow

FRONTEND
React + Vite
Dashboard, chat, calendar

→

API LAYER
Vercel functions
Same-origin server proxy

→

WORKFLOWS
n8n Cloud
Assignments + AI agent

02 REQUEST FLOWS

ASSIGNMENTS READ

1. Browser requests GET /api/assignments.
2. Vercel calls the n8n read webhook.
3. The returned assignments array drives cards, counts, calendar and alerts.

AI CHAT + UPDATE

1. Browser sends chat text + session ID to POST /api/chat.
2. Vercel forwards it to the n8n Chat Trigger.
3. After a successful reply, assignments are fetched again.

03 WHAT THE USER SEES

Live updates: assignments load on opening, refresh every 60 seconds while the tab is visible, and refresh after AI replies or a manual retry.

Useful views: statistics, assignment cards, calendar deadlines, study priorities and alerts for upcoming or overdue work.

Resilience: loading, empty, error and retry states; no fake assignment records.

04 DEPLOYMENT NOTES

Vercel hosts the built frontend and the two serverless API routes. N8N_ASSIGNMENTS_URL is a server-only environment variable. Chat session ID and visible conversation history are stored in this browser. Assignment creation or updates depend on the connected n8n workflow.